

Reviewer Pack — Algebraic K-Theory Rank of Graphical Realizations Bounds DPL...

Entry 98d7145089df · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Algebraic K-Theory Rank of Graphical Realizations Bounds DPLL Search Tree Height

Entry ID: 98d7145089df **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-29 00:00:52 UTC

1. Conjecture

Field A (mathematical branch): Algebraic K-theory **Field B** (complexity object): Complexity Theory: DPLL Search Tree Complexity

Statement:

[‘For any k-CNF instance with n variables, the rank of the algebraic K-theory group of its graphical realization is upper-bounded by a function $f(n)$ that depends only on the number of variables.’, ‘This function $f(n)$ satisfies the following: for all instances with $n \leq 40$, $E[\text{DPLL_search_tree_height}] = O(f(n))$.’, ‘For any counterexample where this bound does not hold, there exists an algebraic K-theory group rank that is arbitrarily large.’]

Rationale (proposer’s reasoning):

[‘Algebraic K-theory provides a framework to study the properties of rings and modules. Graphical realizations of CNFs have been used in studying resolution complexity. By relating the K-theory rank of these graphs to the DPLL search tree height, we may uncover new structural insights into NP-completeness.’, ‘Previous work has shown that algebraic structures can be used to bound computational complexities. This conjecture aims to extend this idea by connecting algebraic K-theory with the complexity of solving satisfiability problems.’]

Taxonomy category: ALGEBRAIC_K_THEORY-bound_DPLL_SEARCH_TREE_HEIGHT (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 629fd34e5b7c1558

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if for all k-CNF instances with $n \leq 40$ variables, the empirical mean of DPLL search tree heights across 30 random seeds is within $O(f(n))$ as calculated by a chi-squared test with $p\text{-value} \geq 0.05$. The conjecture is falsified if any seed produces a

DPLL search tree height outside of $O(f(n))$ with a confidence interval that does not overlap with the expected value.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
KARP_LIPTON	SAFE	1.00	SAFE	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Algebraic K-theory" AND "DPLL search tree height" - "graphical realization" AND "algebraic K-theory rank" AND "CNF instance" - "function $f(n)$ " AND " $E[\text{DPLL_search_tree_height}] = O(f(n))$ " AND "number of variables"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def gaussian_elimination(matrix):
    rows = len(matrix)
    cols = len(matrix[0])
    rref_matrix = [row[:] for row in matrix]

    for i in range(rows):
        # Find pivot
        pivot_row = i
        for j in range(i, rows):
            if abs(rref_matrix[j][i]) > abs(rref_matrix[pivot_row][i]):
                pivot_row = j

        # Swap rows
        rref_matrix[i], rref_matrix[pivot_row] = rref_matrix[pivot_row], rref_matrix[i]

        # Eliminate below the pivot
        for j in range(i + 1, rows):
            factor = -rref_matrix[j][i] / rref_matrix[i][i]
            for k in range(cols):
                if rref_matrix[i][k] != 0:
                    rref_matrix[j][k] += factor * rref_matrix[i][k]
```

```

# Eliminate above the pivot
for i in range(rows - 1, -1, -1):
    factor = 1 / rref_matrix[i][i]
    for j in range(cols):
        if rref_matrix[i][j] != 0:
            rref_matrix[i][j] *= factor

return rref_matrix

def rank(matrix):
    rref_matrix = gaussian_elimination(matrix)
    rank = sum(1 for row in rref_matrix if any(row))
    return rank

def generate_k_cnf(n, m):
    variables = list(range(n))
    clauses = []
    for _ in range(m):
        clause = random.sample(variables, 3)
        clause.append(-random.choice(clause))
        clauses.append(clause)
    return clauses

def dpll_search_tree_height(clauses):
    def dpll(model, clauses):
        if not clauses:
            return True
        unit_clause = next((c for c in clauses if len(c) == 1), None)
        if unit_clause:
            literal = unit_clause[0]
            model[literal] = True
            new_clauses = [c for c in clauses if literal not in c and -literal not in c]
            return dpll(model, new_clauses)
        pure_literal = next((l for l in variables if (l in model and not model[l]) or (-l in model and not model[-l])), None)
        if pure_literal:
            model[pure_literal] = True
            new_clauses = [c for c in clauses if pure_literal not in c and -pure_literal not in c]
            return dpll(model, new_clauses)
        literal = random.choice(variables)
        model[literal] = True
        new_clauses = [c for c in clauses if literal not in c and -literal not in c]
        if dpll(model, new_clauses):
            return True
        del model[literal]
        model[-literal] = True
        new_clauses = [c for c in clauses if -literal not in c and literal not in c]
        return dpll(model, new_clauses)

    variables = list(range(1, n + 1))
    model = {}
    return len(variables) if dpll(model, clauses) else float('inf')

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n_values = [5, 10, 15, 20, 30, 40]
    total_heights = 0

```

```

instances_tested = 0

for n in n_values:
    for _ in range(5): # Ensure at least 30 instances per seed
        clauses = generate_k_cnf(n, int(1.2 * n))
        graphical_realization = [[abs(lit) for lit in clause] for clause in clauses]
        k_theory_rank = rank(graphical_realization)
        height = dpll_search_tree_height(clauses)
        total_heights += height
        instances_tested += 1

mean_height = total_heights / instances_tested
conjecture_holds = True
counterexample = ""

if mean_height > n_values[-1] * 2: # Arbitrary upper bound for demonstration
    conjecture_holds = False
    counterexample = "mean_height_exceeds_upper_bound"

return {
    "metric_name": "DPLL Search Tree Height",
    "metric_value": mean_height,
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if sys.argv[1:] else [2, 3, 5, 7, 11, 13, 17, 19, 23, 29]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {{\"seed\": {seed}, **result}}")
        results.append(result)

    mean_height = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_height} std=0.0 support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(s for s, r in zip(seeds, results) if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{r['counterexample']}\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE reason=unknown")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

```

Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_aac6c988.py", line 129, in <module>
    result = run_trial(seed)
              ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_aac6c988.py", line 102, in run_trial
    k_theory_rank = rank(graphical_realization)
                     ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_aac6c988.py", line 50, in rank
    rref_matrix = gaussian_elimination(matrix)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_aac6c988.py", line 27, in gaussian_elim
    if abs(rref_matrix[j][i]) > abs(rref_matrix[pivot_row][i]):
        ~~~~~^
IndexError: list index out of range

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing data, which means that the empirical mean of DPLL search tree heights could not be calculated and compared to the expected value $O(f(n))$. Therefore, the support condition for the conjecture cannot be unambiguously met. | next: Re-run the test without errors to verify if the conjecture is supported or falsified.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12313
2	propose	ollama_remote	glm4:latest	0	0	11434
3	preregistration	ollama_remote	glm4:latest	0	0	6282
4	novelty	ollama_remote	glm4:latest	0	0	4833
5	novelty	ollama_remote	glm4:latest	0	0	5633
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	45579
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10010
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14034
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16163
10	judge	ollama_remote	glm4:latest	0	0	16792

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 143074 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/98d7145089df.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/98d7145089df.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/98d7145089df.tar.gz` (if generated)